Sprint 4 Plan, Parking Finder, Parking Finder, 3/10/26, Revision 1, 2/24/26

Goals:

- Establish parking lot zones on UCSC campus
- Allow to automatically mark another person's marker as "taken"
- Polish the UI
- Refactor the codebase
- Try to implement TA suggestion as best as we can

User Story 4.1 (13 pts)

As a driver, I want to press a button that checks for nearby parking spot markers and marks one as taken, or gives me the option to create a new "taken" marker at my current location if none are nearby, so that I can quickly keep parking availability accurate.

Tasks:

- Design the button placement and user flow in the map screen UI (1.5 hrs)
- Implement logic to search for nearby active markers within a defined radius of the user's current location (2 hrs)
- Add rule to select only one marker if multiple nearby markers are found (such as closest marker) (1.5 hrs)
- Update the selected marker's status to "TAKEN" in Firebase and refresh the map (2 hrs) If no nearby marker is found, show a prompt asking the user whether they want to create a new "taken" marker at their current location (1.5 hrs)
- Implement creation of a new "taken" marker at the user's location when confirmed (2 hrs)

Estimated Time: 10.5 hrs

User Story 4.2 (8 pts)

As a driver, I want to manually mark a parking spot as taken so that I can update the map when I notice a spot is no longer available.

Tasks:

- Add "Mark as Taken" action to spot detail modal/callout (2 hrs)

- Restrict action to valid spots (open, not expired) (1.5 hrs)
- Update Firestore document status to “TAKEN” on user action (2 hrs)
- Sync marker update in real time on the map (1.5 hrs)
- Test manual marking across multiple users/devices (1 hr)

Estimated Time: 8 hrs

User Story 4.3 (8 pts)
As a driver, I want the app to automatically mark only one nearby parking spot as taken so that multiple spots in the same area are not incorrectly closed at once. (4 markers in one area -> should only mark one spot as taken automatically)

Tasks:

- Define selection rule for which nearby spot is eligible (e.g., closest, oldest) (1.5 hrs)
- Modify auto-take logic to select a single target spot (2 hrs)
- Add lock/flag to prevent multiple auto-takes within the same radius/time window (2 hrs)
- Test behavior with multiple clustered markers (1.5 hrs)
- Add logging/debug output for verification (1 hr)

Estimated Time: 8 hrs

User Story 4.4 (13 pts)
As a driver, I want to only be able to drop parking markers within parking lots and streets so that reports remain accurate and relevant to valid parking areas.

Tasks:

- Obtain or define geofence boundaries for UCSC parking lots (polygons or bounding boxes) (3 hrs)
- Store parking lot boundaries locally or in Firestore (2 hrs)
- Implement point-in-boundary check when user drops a pin (2 hrs)
- Block invalid drops and display user feedback (error/toast) (2 hrs)
- Visually indicate valid parking zones on the map (overlay/highlight) (2 hrs)

Estimated Time: 11 hrs

Initial Task Assignment:

- Kenny: Refactor codebase
- Ethan: Get started on user story
- Maxwell: Refactor codebase
- Kyle: Bug fix

Team Roles:

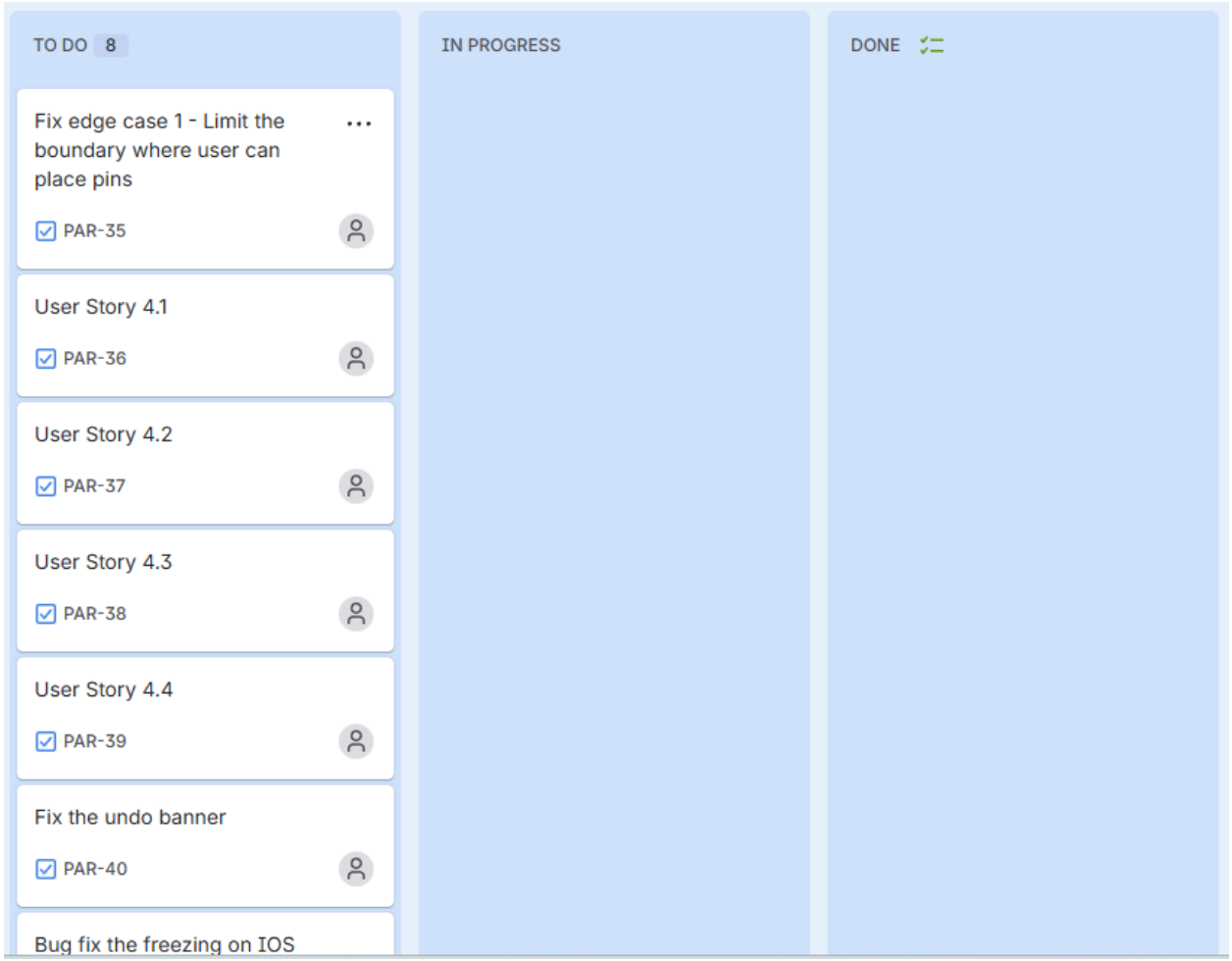
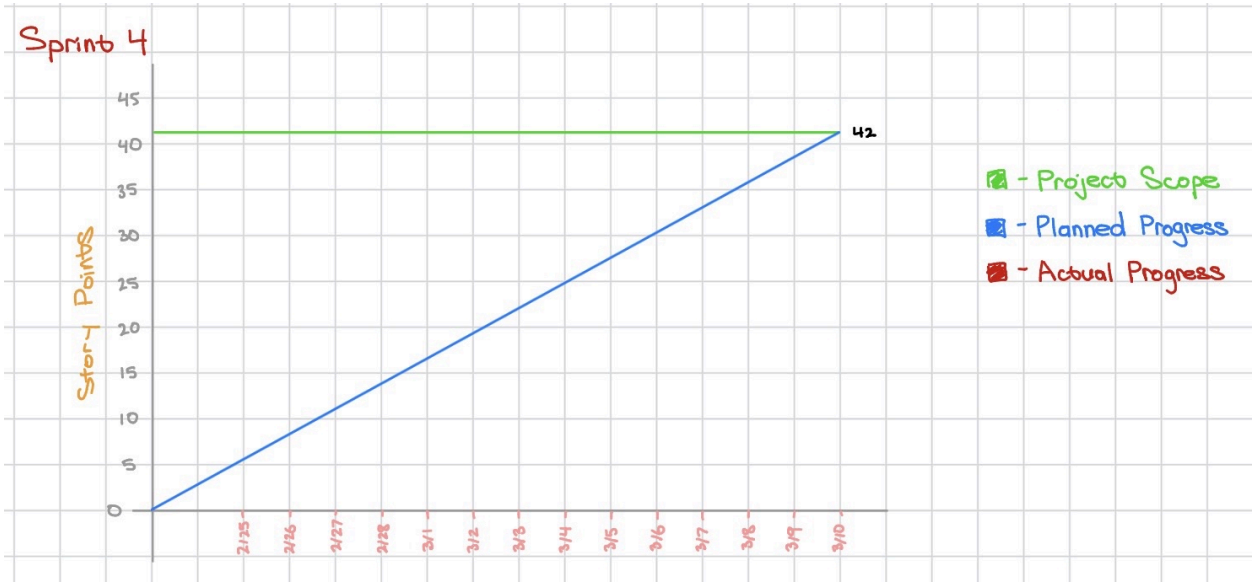
- Kenny: Product Owner & Scrum Master
- Ethan: Developer
- Maxwell: Developer
- Kyle: Developer

Scrum Times:

MWF after class

Initial Scrum Board:

Sprint 4 Plan, Parking Finder, Parking Finder, 3/10/26, Revision 1, 2/24/26



Burnup Chart: